

MAHTA AKHYANI

(+98) 933-254-9264 - Mahta.akhyani@gmail.com - <https://mahtaakhyani.github.io> - <https://github.com/mahtaakhyani>

RESEARCH INTERESTS

- Biomechanics, Bio-inspired Robots, Assistive Technology, Human Robot Interaction, Social Cognitive Robots

PUBLICATIONS

- Submitted **Mahta Akhyani, Hadi Moradi**, "Modular Customizable ROS-Based Framework for Rapid Development of Social Robots" [Github](#) - [arXiv \[PDF\]](#)
- In Prepration **Pegah Soleiman, Mahta Akhyani, Alireza Kargar, Hadi Moradi**, "Empathy toward Robotic Social Environment vs. Human Social Environment, Comparison between people with Autism Spectrum Disorder and typically developing ones"

INTERNSHIPS

- Summer 2024 **Research Assistant, GIST (Gwangju Institute of Science and Technology), South Korea**
- Developing and implementing Trust-POMDP algorithm for human-exoskeleton interactions

EDUCATION

- 2018 - 2023 **Bachelor of Science: Chemical Engineering** (Subdiscipline: Biotechnology)
University of Tehran - Iran
GPA: 15.63/20 (16/20 in the last 2 years)
B.Sc. thesis (**Full Marked**) : **Design of mobile robot with SLAM capabilities for pipe inspection in chemical and petroleum industries** [Media](#) - [arXiv \[PDF\]](#)

SKILLS

Proficient:

Python - ROS - Gazebo - Javascript - HTML/CSS - Robotics platforms (e.g. Raspberry Pi, Arduino, Jetson Nano) - Version control (Git) - 3D modeling (Solidworks)

Experienced:

Django REST framework - Ubuntu - Machine Learning and AI (TensorFlow, Pandas, NumPy, Matplotlib)

Intermediate:

C++, C, AVR, Computer vision and AI (OpenCV, MediaPipe, Praat)

RESEARCH EXPERIENCE

- 08/2022 - 10/2023 **[Terahertz Photoelectronics Lab](#)**
Department of Computer and Electrical Engineering, University of Tehran
Supervisor: Prof. Mohammad Neshat
Project Title: **Electrically Conductive Mixed Matrix Membrane-Based 3-Layered Geometrically Matching Head Phantom** [More info](#)
- Synthesized and characterized polymers including PANI, PDMS, starch, RTV-2, graphene and carbon black for a multilayer head phantom through CPC and MMM fabrication
- 04/2022 - 07/2022 **B.Sc. Thesis Project** [Media](#) - [arXiv \[PDF\]](#)
Design of mobile robot with SLAM capabilities for pipe inspection in chemical and petroleum industries
- Development of a sensor fusion pipeline integrating Kinect camera and IMU data to improve odometry estimates
 - Simulated and built a mobile robot platform and testing the ROS/Gmapping SLAM approach on it in a maze
- 06/2021 - 07/2024 **[Advanced Robotics and Intelligent Systems Lab](#)**
Department of Computer and Electrical Engineering, University of Tehran
Supervisor: Prof. Hadi Moradi
Project Title: **Understanding Empathy Towards Social Robots and Its Implications for Individuals with Autism**
- Investigated empathy in human-robot and robot-robot interactions using EEG, comparing responses of individuals with autism and neurotypical participants
 - Examined joint attention responses in autism for both robotic and human social interactions
- Project Title: **Modular Customizable ROS-Based Framework for Rapid Development of Social Robots**
- Developed reusable robot skill modules (face analysis, speech processing, dialogue agents, actuator control)
 - Engineered a distributed data processing pipeline optimized for multiple sensors

- Utilized video/photo editing to design facial expressions for the robot

03/2019 - 02/2020 [Pharmacology Lab](#)

Department of Chemical Engineering, University of Tehran

Supervisor: Prof. Reza Zarghami

Project Title: **Chem-E-Car Competition** (Designed and constructed a car powered by synthesized chemical energy source and brake system)

TEACHING EXPERIENCE

Spring 2022 **Teaching Assistant**

Advanced Robotics (Graduate)

Department of computer and electrical engineering, University of Tehran

Lecturer: Prof. Hadi Moradi

- Provided hands-on tutoring to students on ROS-based localization techniques for Anki Vector robot

Winter 2021 **Teaching Assistant**

Learning Python with Computer Vision approach

[RoboTech Academy](#)

- Developed assignments and quizzes, and provided tutoring support to reinforce learning concepts

Summer 2019 **Self-Directed Volunteer Teacher**

Teaching Python fundamentals during summer to 12 students

JOB EXPERIENCE

04/2020 - 02/2021 **Co-founder and Leader of Fanjoo**

Online platform for peer knowledge exchange and optional income

Addressed gap between education and workforce needs by empowering users to gain in-demand technical skills outside traditional academia. [More info](#) - [GitHub](#)

Skills achieved:

- Effective Collaboration
- Self Disciplinary
- Project Management
- Team Leadership

10/2019 - 01/2020 **Django Backend Developer**

UTech Academy - Tehran, Iran

- Independent back-end Django REST developer [GitHub](#)

CERTIFICATIONS AND EXTRA COURSES

April 2022 **TOEFL Test**

- Overall Score: **97/120** (r27,l25,s22,w23)

Spring 2021 **Fundamentals of Mechatronics (Graduate-level)**

Department of electrical and computer engineering, University of Tehran, Iran

- Project Grade: **120 out of 80** (Extra Credit)
- Overall Grade: **19.5 out of 20**

Winter 2020 **Deep Learning and Computer Vision**

[RoboTech Academy](#)

Summer 2019 **Back-End Developing with Django**

UTech Academy

- Grade: **100/100**

Summer 2018 **Programming with Python**

[Tehran Institute of Technology](#)

- Grade: **95/100**

HONOURS AND AWARDS

Summer 2018 **University Entrance Exam top 0.7%**

Ranked top 0.7% in among 144000 participants in [IUEE](#) for B.Sc.

Spring 2017 **Best Provincial Project, 18th Khwarizmi Scientific Competition Youth Awards ([KYA](#)), Tehran**

Project Title: "Designing Different Types of Electrical Compass".

REFERENCES

Professor **Prof. Hadi Moradi** moradih@ut.ac.ir

Assistant Professor **Prof. Mohammad Neshat** mneshat@ut.ac.ir

Lecturer **Prof. Kambiz Shoarinejad** kambiz1@ucla.edu